

CHAPTER 4

Waves

From music to sunlight to Wi-Fi, waves move energy all around you every second.



Easy Read Edition

WHAT WE WILL LEARN

- ★ What are waves?
- ★ Two main types of waves
- ★ Parts of a wave
- ★ Wave energy
- ★ The light spectrum
- ★ Sound waves
- ★ What waves do
- ★ Lenses and light

4.1 What Are Waves?

Throw a rock in a pond. Ripples spread out in circles. A leaf in the water just bobs up and down.

Here's the cool part: the leaf **doesn't move** across the pond. The ripples move. The water mostly stays put.

That is a wave!

A **wave** moves energy from one place to another. It does NOT carry the stuff with it.

Stadium Wave

Think of a stadium "wave." Each person stands up then sits down. The wave goes around the stadium. But nobody walks anywhere!

That's how all waves work. The pattern moves. The stuff stays.

4.2 Two Types of Waves

All waves are one of two kinds.

1. Mechanical Waves

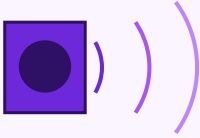
These need something to travel through. They need air, water, or some material. Sound is one. So are ocean waves.

2. Electromagnetic (EM) Waves

These can travel through empty space! They don't need air or anything. Light is one. So are radio waves and microwaves.

MECHANICAL

needs air/water/matter



sound from
speaker travels
through air

ELECTROMAGNETIC

can travel through space!



sun → Earth through empty space

Mechanical waves need stuff. EM waves don't!

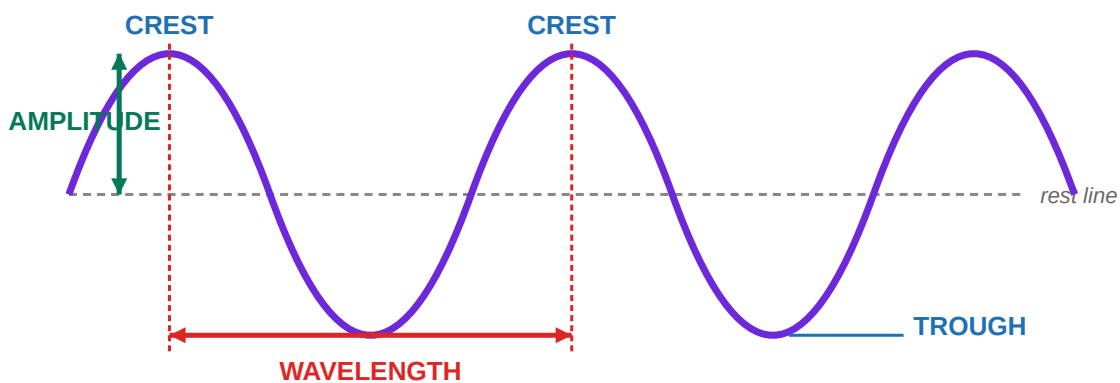


Cool Fact

In space, no one can hear you scream. There is no air for sound to travel through! But you can still SEE things in space, because light is EM and doesn't need air.

4.3 Parts of a Wave

Waves have parts you need to know.



Parts of a wave.

Crest

The TOP of a wave.

Trough

The BOTTOM of a wave.

Wavelength

Distance from one crest to the next crest.

Amplitude

How tall the wave is. From rest line to the crest.

Frequency

How many waves pass by per second. Measured in **Hertz (Hz)**.

4.4 Wave Energy

Two things tell you how much energy a wave has:

Bigger Amplitude = More Energy

Tall waves carry more energy. A tsunami has WAY more energy than a tiny ripple.

Higher Frequency = More Energy

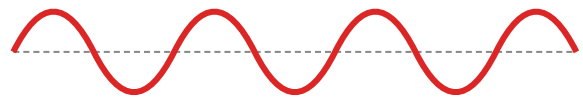
Lots of waves per second = lots of energy. X-rays have more energy than radio waves.

LOW ENERGY



small & slow waves
like: quiet sound

HIGH ENERGY



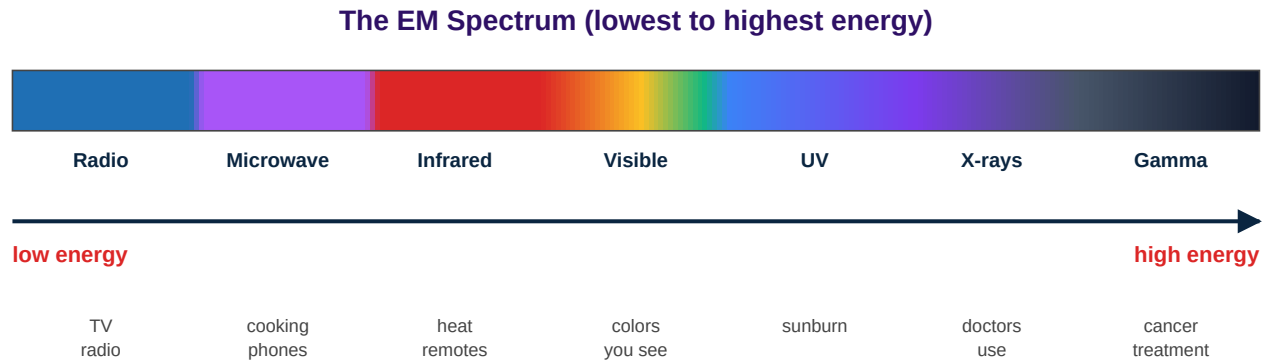
big & fast waves
like: loud sound

Bigger and faster waves carry more energy.

4.5 The EM Spectrum

The light you see is just a small part of a HUGE family of EM waves.

The whole family is called the **electromagnetic spectrum**.



Visible Light = Red, Orange, Yellow, Green, Blue, Purple
(the only ones you can see with your eyes!)

The EM Spectrum. Visible light is just a tiny part!

★ Phone Power

Your phone uses LOTS of these waves!

- Wi-Fi and cell signal = microwaves
- Screen light = visible light
- Face ID = infrared
- GPS = microwaves from satellites

4.6 Sound Waves

Sound is a mechanical wave. It needs air (or water, or something) to move through.

When something vibrates, it pushes the air around it. Those pushes travel as a wave. When they reach your ear, you hear a sound!

Pitch (how high or low)

Pitch depends on FREQUENCY. High frequency = high pitch. Low frequency = low pitch.

Volume (how loud)

Volume depends on AMPLITUDE. Big amplitude = loud. Small amplitude = quiet.



Mouse squeak
HIGH pitch



Elephant rumble
LOW pitch



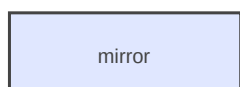
Concert
LOUD volume

4.7 What Waves Do

When a wave hits something, five things can happen.

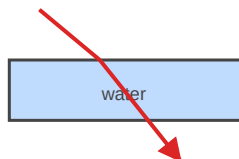
Behavior	What Happens	Example
Reflection	Wave bounces back	Looking in a mirror
Refraction	Wave bends	A straw looks bent in water
Absorption	Wave is soaked up	Black shirt gets hot in sun
Diffraction	Wave bends around corners	Hearing through a doorway
Transmission	Wave passes through	Light through a window

Reflection



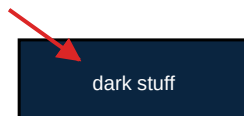
bounces back

Refraction



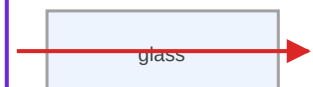
bends!

Absorption



soaked up

Transmission



goes through

Four of the ways waves act when they hit something.

4.8 Lenses and Light

A **lens** is curved glass or plastic that bends light. There are two types:

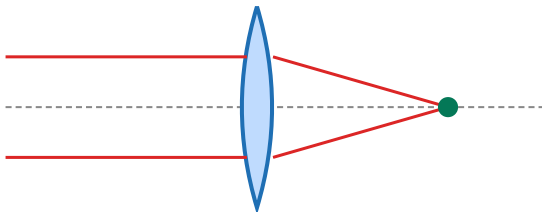
Convex Lens (fat in the middle)

Brings light together. Used in magnifying glasses, cameras, and your eyes!

Concave Lens (thin in the middle)

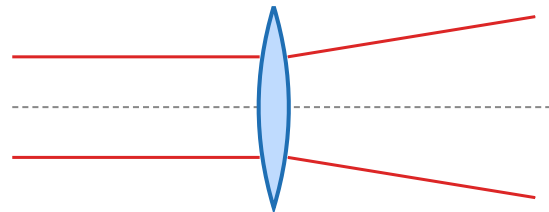
Spreads light apart. Used in some glasses to fix near-sighted vision.

CONVEX (brings together)



like a magnifying glass

CONCAVE (spreads apart)



like glasses for near-sight

Two kinds of lenses. They bend light in different ways.



Did You Know?

Your eyes have CONVEX lenses inside! They bend light to form a picture on the back of your eye.



Check Yourself

1. What's the main difference between a mechanical wave and an EM wave?
2. What part of a sound wave makes it loud?
3. What kind of lens is in a magnifying glass?

★ CHAPTER 4 — BIG IDEAS

- A **wave** moves energy from one place to another. It doesn't move the stuff.
- **Mechanical waves** (sound) need a material to travel through.
- **EM waves** (light) can travel through empty space.
- Wave parts: **crest** (top), **trough** (bottom), **wavelength**, **amplitude**, **frequency**.
- Bigger amplitude or higher frequency = more energy.
- The **EM spectrum** goes from radio (low energy) to gamma rays (high energy).
- For sound: **pitch** = frequency, **volume** = amplitude.
- Waves can **reflect**, **refract**, **absorb**, **diffract**, or **transmit**.
- **Convex lenses** bring light together. **Concave lenses** spread it apart.